Algebra 2 Teacher Answer Key

Partner A and Partner B answers are identical for each numbered problem.

Unit 1: Functions & Transformations

#	1.1 Function Operations & Evaluation	1.2 Transformations of Functions
1	10	$y = (x-3)^2 + 2$
2	19	$g(x) = f(x-2) + 4$
3	$x \geq 4$	$y = -2x - 3$
4	$f^{-1}(x) = \dfrac{x+6}{3}$	$9x^2$
5	$x^2 + 5x - 5$	$y = (x+4)^2 - 1$
6	15	$y \geq -3$
7	3	odd
8	5	11

Unit 2: Quadratics & Complex Numbers

#	2.1 Solving Quadratic Equations	2.2 Complex Numbers & Parabolas
1	$x = 4,\ 5$	$-i$
2	$x = 1,\ -7$	$4 - 3i$
3	$x = 2 \pm \sqrt{3}$	$5 + i$
4	$x = -1 \pm 2i$	$-1 + 3i$
5	-31	5
6	$(3, 2)$	$y = x^2 - 2x - 3$
7	sum 5, product 6	$x = \pm 3i$
8	$t = 4$ s	4

Unit 3: Polynomial Functions

#	3.1 Polynomial Operations & Graphs	3.2 Zeros, Factoring & Inequalities
1	$x^3 - x^2 - 5x + 2$	$(x-3)(x^2+3x+9)$
2	$x^2 - 2x - 3$	$x = -3,\ -2,\ 2$
3	-3	$x = 1$
4	$x(x-2)(x+2)$	$x^3 - 4x^2 + x + 6$
5	$x = 1,\ 2,\ 3$	$x = \pm 2,\ \pm 2i$
6	both ends down	5
7	4	$3 - i$
8	$y = (x+2)(x-1)(x-3)$	$x < -3$ or $x > 1$

Unit 4: Radicals & Rational Exponents

#	4.1 Simplifying Radicals & Exponents	4.2 Radical Equations & Functions
1	$6\sqrt{2}$	$x = 9$
2	9	$x = 3$
3	$3\sqrt{10}$	$x = 13$
4	$5\sqrt{5}$	$x = 4$
5	$2\sqrt{3}$	$x = 4$
6	$\dfrac{3+\sqrt{5}}{2}$	4
7	$x^{3/2}$	$x \geq 1$
8	$x^{4/3}$	60 mph

Unit 5: Rational Functions

#	5.1 Rational Expressions & Equations	5.2 Graphs of Rational Functions
1	$\dfrac{x+3}{x+2}$	$x = 3$
2	$\dfrac{1}{x}$	$y = 3$
3	$x^2 + 2x$	$(2, 4)$
4	$\dfrac{5x-3}{x(x-1)}$	-2
5	$x = 3$	$x \neq \pm 3$
6	no solution	$y = \dfrac{1}{x-2} + 1$
7	2 hours	$y = x$
8	8	$\dfrac{9}{2}$

Unit 6: Exponential & Logarithmic Functions

#	6.1 Exponential & Log Basics	6.2 Exp & Log Equations and Graphs
1	5	1.86
2	3	$\log(x^2y)$
3	6	10
4	2	e^3
5	8	4
6	5	$y = 3 \cdot 2^x$
7	1,600	$y = -3$
8	\$1,102.50	4

Unit 7: Sequences & Series

#	7.1 Arithmetic & Geometric Sequences	7.2 Series, Binomials & Applications
1	41	35
2	$a_n = 3n + 4$	21
3	384	765
4	14	4
5	210	10
6	126	$x^3 + 6x^2 + 12x + 8$
7	16	51.2 ft
8	14	6

Unit 8: Trigonometry

#	8.1 Radians, Unit Circle & Right-Triangle Trig	8.2 Graphs of Trig Functions & Trig Equations
1	$\frac{5\pi}{6}$	3
2	2π	π
3	$\frac{1}{2}$	$y = 4$
4	$-\frac{1}{2}$	$y = 3\sin x + 2$
5	-1	$x = \frac{\pi}{6}, \frac{5\pi}{6}$
6	$\frac{4}{5}$	$x = \frac{2\pi}{3}, \frac{4\pi}{3}$
7	$30°$	1
8	$40°$	45 ft